

Final: Best Model

Team Name: Deep Learners

Teammates Involved:

John Nguyen- W1419143@student.hccs.edu

Joseph Armijo- W216991967@student.hccs.edu

Misty Richardson- W4626687@student.hccs.edu

Saima Sano- W216790818@student.hccs.edu

Willie Breaux- W213801233@student.hccs.edu

FINAL REGRESSION COMPARISON TABLE:

	Model	MAE	MSE	R ²
0	Linear Regression	325.635833	1.552248e+06	0.948265
1	Decision Tree Regressor	162.005558	1.136256e+06	0.962130
2	Random Forest Regressor	160.055764	1.004542e+06	0.966520
3	Gradient Boosting Regressor	338.445630	1.182456e+06	0.960590
4	K-Nearest Neighbors Regressor	198.635652	1.965696e+06	0.934485
5	Voting Ensemble (top 3)	249.671497	1.008389e+06	0.966391
6	Average Ensemble	223.252492	1.099541e+06	0.963353
7	Bayesian Ridge Regressor	283.619572	1.250871e+06	0.958310

The best-performing model was the Random Forest Regressor because it produced the lowest average prediction error, minimized heavy error penalties, and explained the greatest proportion of variance in the target variable. Random Forest performs well because it aggregates multiple decision trees. While the ensemble approach improved overall stability, the Random Forest Regressor delivered accurate, reliable performance across all metrics. Random Forest's ability to reduce error while maintaining strong explanatory power makes it ideal for this ML task.